

HSBC BANK

Personal Loan Acceptance Analysis

Customer Analytics & Marketing Optimization Report

Business Problem Statement

HSBC Bank wants to optimize its personal loan marketing campaigns by identifying customers who are more likely to accept a personal loan offer.

The bank has historical customer data containing demographic details, financial information, banking product usage, and previous loan acceptance information.

Your task is to analyse the dataset and answer key business questions such as:

- Which type of customers are more likely to accept a personal loan?
- What customer characteristics influence loan acceptance?
- Which banking services are commonly used by customers who accepted loans?
- How can HSBC improve campaign targeting to reduce marketing cost and improve conversion rates?

The final objective is to provide actionable business recommendations for the marketing team.

1. Objective of the Report

HSBC has been contacting a large number of customers with personal loan offers, but very few actually accept. This makes campaigns expensive and inefficient.

This report studies 5,000 customer records to:

- Find out which customers are most likely to accept a loan.
- Show which factors matter most (income, education, banking relationship, spending, etc.).
- Recommend a simpler, more focused targeting strategy for future campaigns.

2. Procedure

The analysis followed six simple steps:

Step 1: Data Understanding — Reviewed all columns and confirmed “Personal Loan” as the target variable.

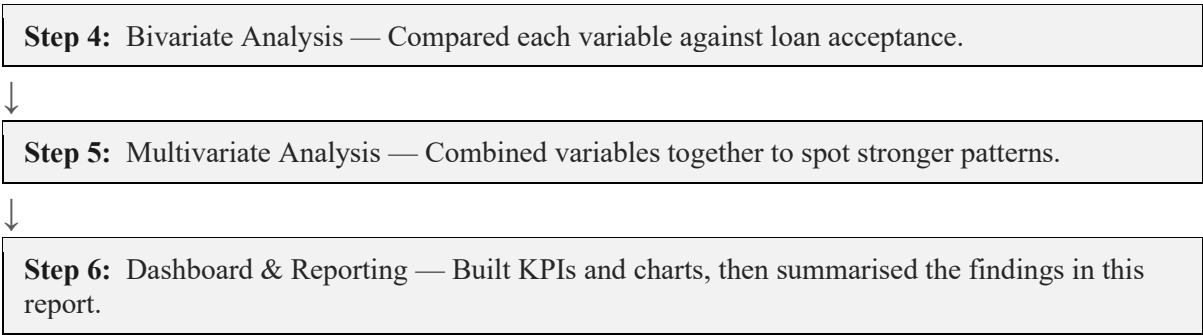


Step 2: Data Cleaning — Checked for missing values, duplicates, and invalid entries.



Step 3: Univariate Analysis — Looked at each variable on its own (age, income, education, family size).





3. Dataset

Key numbers from the cleaned dataset:

Metric	Value
Total customer records	5,000
Total attributes (columns)	15 original + derived grouping fields
Target variable	Personal Loan (0 = No, 1 = Yes)
Customers who accepted the loan	480
Customers who did not accept the loan	4,520
Overall loan acceptance rate	9.60%
Average customer age	45.3 years
Average annual income	₹73.8K
Average monthly credit card spend (CCAvg)	₹1.94K
Average mortgage value	₹56.5K
Average family size	2.4 members

Only 9.60% of customers accept the loan. The sections below show which customers make up most of that 9.60%.

4. Business Insights

4.1 Income Is the Strongest Predictor of Loan Acceptance

Acceptance rises sharply as income increases:

Income Group	Customers	Accepted	Acceptance Rate
Below ₹25K	1,869	0	0.00%
₹50K – ₹74K	1,042	6	0.58%
₹75K – ₹99K	867	35	4.04%
₹100K – ₹149K	769	219	28.48%

Income Group	Customers	Accepted	Acceptance Rate
₹150K – ₹199K	434	217	50.00%
₹200K and above	19	3	15.79%

Customers earning above ₹100K/year are 5–10 times more likely to accept than those earning below ₹75K. Accepting customers earn ₹144.75K on average, versus ₹66.24K for non-acceptors.

4.2 Education Level Influences Acceptance, But Modestly Compared to Income

Education Level	Customers	Accepted	Acceptance Rate
Undergraduate	2,096	93	4.44%
Graduate	1,403	182	12.97%
Professional	1,501	205	13.66%

Graduates and Professionals accept the loan about 3 times more often than Undergraduates.

4.3 Existing Term Deposit (TD) Relationship Is a Major Cross-Sell Signal

TD Account Holder	Customers	Accepted	Acceptance Rate
No	4,698	340	7.24%
Yes	302	140	46.36%

TD account holders accept loans at 46.36% — nearly 6.5 times higher than non-holders. This is the strongest single indicator in the dataset.

4.4 Credit Card Spending Behaviour Is a Strong Early-Warning Indicator

Avg. Monthly Card Spend (CCAvg, ₹K)	Customers	Accepted	Acceptance Rate
0 – 1	1,683	47	2.79%
1 – 2	1,376	45	3.27%
2 – 3	1,039	56	5.39%
3 – 4	319	104	32.60%
4 – 5	219	81	36.99%
5 and above	364	147	40.38%

Acceptance jumps once monthly card spend crosses ₹3K, rising from single digits to 30–40%. Acceptors spend ₹3.91K/month on average, versus ₹1.73K for non-acceptors.

4.5 Mortgage Holders Convert at a Much Higher Rate

Mortgage Value	Customers	Accepted	Acceptance Rate
No Mortgage	3,462	312	9.01%
Low (₹1–100K)	282	13	4.61%
Medium (₹101–200K)	1,055	79	7.49%

Mortgage Value	Customers	Accepted	Acceptance Rate
High (₹200K+)	201	76	37.81%

Customers with mortgages above ₹200K accept loans at 37.81%, well above average. They are already comfortable with larger credit obligations.

4.6 Age and Work Experience Have Only a Marginal Effect

Age Group	Acceptance Rate	Experience Group	Acceptance Rate
20–29	10.04%	0–10 yrs	10.76%
30–39	10.18%	11–20 yrs	8.83%
40–49	9.31%	21–30 yrs	9.30%
50–59	8.85%	30+ yrs	9.59%
60+	10.24%	—	—

Acceptance stays in a narrow 8.8%–10.8% range across all age and experience groups. Neither factor is useful for targeting on its own.

4.7 Family Size Has a Small but Noticeable Effect

Family Members	Customers	Accepted	Acceptance Rate
1	1,464	106	7.24%
2	1,310	108	8.24%
3	1,009	133	13.18%
4	1,217	133	10.93%

Customers with 3–4 family members accept loans slightly more often (11–13%) than 1–2 person households (7–8%), likely due to higher household expenses. The effect is small.

4.8 Digital Banking and Credit Card Ownership Are Weak Standalone Predictors

Variable	Segment	Customers	Acceptance Rate
Online Banking	Non-user	2,016	9.38%
Online Banking	User	2,984	9.75%
Credit Card (HSBC)	Non-holder	3,530	9.55%
Credit Card (HSBC)	Holder	1,470	9.73%
Securities Account	No	4,478	9.38%
Securities Account	Yes	522	11.49%

These three factors each shift acceptance by less than 2 percentage points. They are not useful for targeting on their own, but can help decide how to deliver an offer once a good customer is identified.

4.9 Combined Profile of a Likely Acceptor

The customer most likely to accept a loan looks like this:

- Annual income above ₹100K (ideally ₹150K–199K, where acceptance peaks at 50%)
- Graduate or Professional education level
- Already holds a Term Deposit (TD) account with HSBC
- Average monthly credit card spend above ₹3K
- Carries a mortgage above ₹200K, or has 3–4 family members

This group is a small part of the current outreach list, but converts at several times the average rate. It should be the core target segment.

4.10 Overall Campaign Efficiency Implication

Over 90% of customers contacted today do not convert. Shifting focus toward the profile in Section 4.9 should raise conversion rates without spending more.

5. Dashboard Summary (Key Performance Indicators)

The Excel dashboard summarises these KPIs, with filters for Income Group, Education, TD Account, and Card Spend:

KPI	Value
Total customers	5,000
Total loan acceptances	480
Overall acceptance rate	9.60%
Highest-converting income band	₹150K–199K (50.00%)
Highest-converting education level	Professional (13.66%)
Highest-converting relationship signal	TD Account holders (46.36%)
Highest-converting spend band	CCAvg ₹5K+ (40.38%)
Weakest predictors	Online banking, credit card ownership, age, experience

6. Strategic Recommendations

6.1 Prioritise High-Income Customers

Focus outreach on customers earning above ₹100K/year, especially the ₹150K–199K group, which converts at 50%. Impact: higher conversion, lower cost per acquisition.

6.2 Cross-Sell to Existing Term Deposit Customers First

TD account holders convert at nearly 5 times the average rate. Make them the first call list, with pre-approved offers. Impact: the most efficient segment to target, at low extra cost.

6.3 Use Credit Card Spend as a Real-Time Targeting Signal

Flag customers spending above ₹3K/month on cards for proactive offers. Impact: allows real-time targeting instead of fixed demographic lists.

6.4 Design Segment-Specific Messaging for Graduates and Professionals

Design offers and messaging for Graduate and Professional customers, who account for most acceptances.

Impact: better response rates and product fit.

6.5 Treat Mortgage Value as a Financial-Maturity Filter

Include customers with mortgages above ₹200K in the priority segment. Impact: better retention and lifetime value, alongside higher conversion.

6.6 De-Prioritise Age, Experience, and Digital-Only Signals as Primary Filters

Don't use age, experience, online banking, or card ownership as main targeting filters — they barely affect acceptance. Use them only to fine-tune how an offer is delivered.

6.7 Reduce Outreach Volume to Low-Income, No-Relationship Customers

Customers earning below ₹75K/year with no TD account and low card spend convert at under 5%. Reducing outreach here frees up budget for the high-value segments above.

7. Conclusion

Loan acceptance is not random. It is concentrated among customers with high income, a TD account, high card spend, higher education, and larger mortgages. Age, experience, online banking, and card ownership matter far less.

By targeting this profile, HSBC can raise its conversion rate above the current 9.60%, cut wasted marketing cost, and build stronger relationships with its most valuable customers.